



Digital Integrated Circuit Design

Lecture-8: Systolic Array

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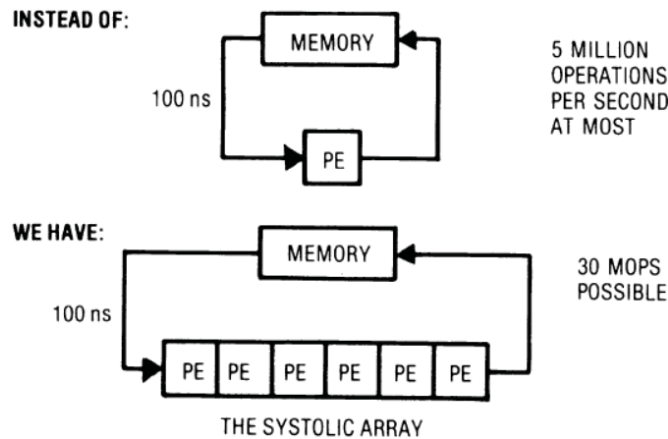
Outline

- ☐ **Systolic System**
 - Processing Element
 - Matrix-Vector
 - Matrix Multiplication
- ☐ Givens Rotation
- ☐ QR Decomposition

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Systolic Array

❑ Basic principle of a systolic system



- Replace single processor with an array of regular processing
- Organize data flow for high throughput with less memory access

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Systolic Array (cont'd)

❑ Matrix-vector multiplication

$$\blacksquare \mathbf{A}_{n \times m}, \mathbf{u}_{m \times 1}, \mathbf{V}_{n \times 1} \rightarrow \mathbf{V}_{n \times 1} = \mathbf{A}_{n \times m} \cdot \mathbf{u}_{m \times 1}$$

$$\rightarrow v_i = \sum_{j=1}^m a_{ij} u_j$$

❑ Matrix-matrix multiplication:

$$\blacksquare \mathbf{A}_{m \times n}, \mathbf{B}_{n \times p}, \mathbf{C}_{m \times p} \rightarrow \mathbf{C}_{m \times p} = \mathbf{A}_{m \times n} \cdot \mathbf{B}_{n \times p}$$

$$\rightarrow c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

❑ Convolution

- $y_n = \sum_{k=-\infty}^{\infty} w_k x_{n-k}; y_n = w_n \otimes x_n;$
- If both w_n and x_n are causal sequences and each of finite length N , i.e., $n = 0, 1, \dots, N-1$.
- $y_n = \sum_{k=0}^{N-1} w_k x_{n-k}$, where $n = 0, 1, \dots, 2N-2$

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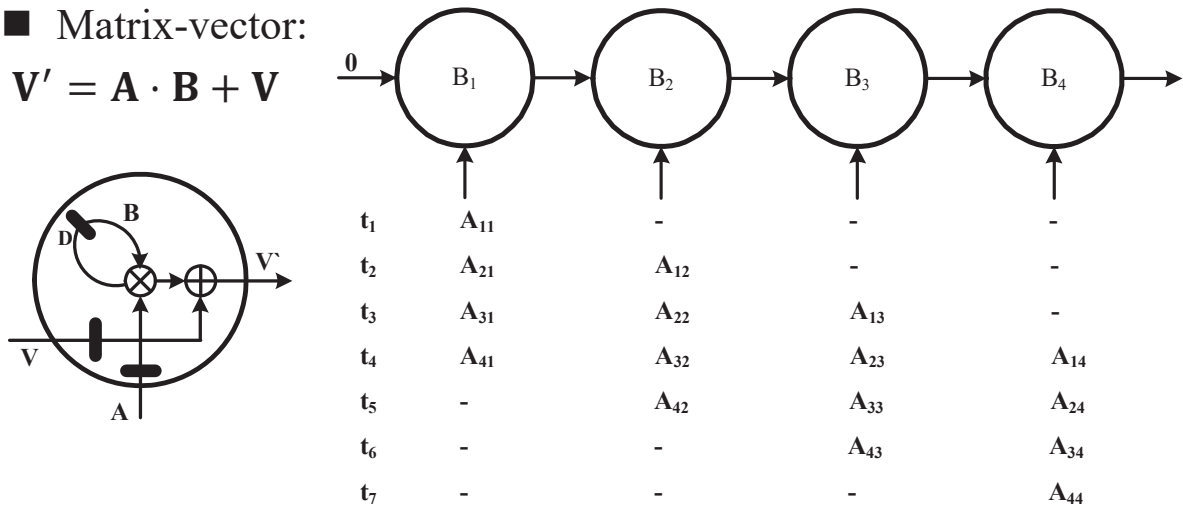
Systolic Array (cont'd)

- In digital filtering, x_n is the input sequences to be processed and w_n is the impulse response function of the processing digital filter.

□ Example

- Matrix-vector:

$$V' = A \cdot B + V$$



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Systolic Array – Matrix Multiplication

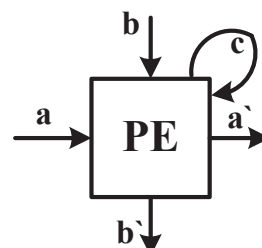
□ 3×3 Matrix multiplication

$$C = A \cdot B = \begin{bmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ a_{20} & a_{21} & a_{22} \end{bmatrix} \cdot \begin{bmatrix} b_{00} & b_{01} & b_{02} \\ b_{10} & b_{11} & b_{12} \\ b_{20} & b_{21} & b_{22} \end{bmatrix}$$

$$= \begin{bmatrix} a_{00}b_{00} + a_{01}b_{10} + a_{02}b_{20} & a_{00}b_{01} + a_{01}b_{11} + a_{02}b_{21} & a_{00}b_{02} + a_{01}b_{12} + a_{02}b_{22} \\ a_{10}b_{00} + a_{11}b_{10} + a_{12}b_{20} & a_{10}b_{01} + a_{11}b_{11} + a_{12}b_{21} & a_{10}b_{02} + a_{11}b_{12} + a_{12}b_{22} \\ a_{20}b_{00} + a_{21}b_{10} + a_{22}b_{20} & a_{20}b_{01} + a_{21}b_{11} + a_{22}b_{21} & a_{20}b_{02} + a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

□ PE for matrix multiplication

- $a' = a$
- $b' = b$
- $c = c + a \cdot b$



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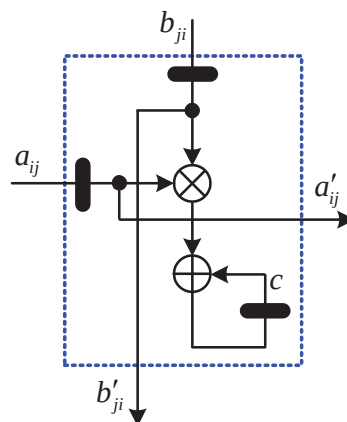
Systolic Array – Matrix Multiplication (cont'd)

■ Architecture PE for matrix multiplication

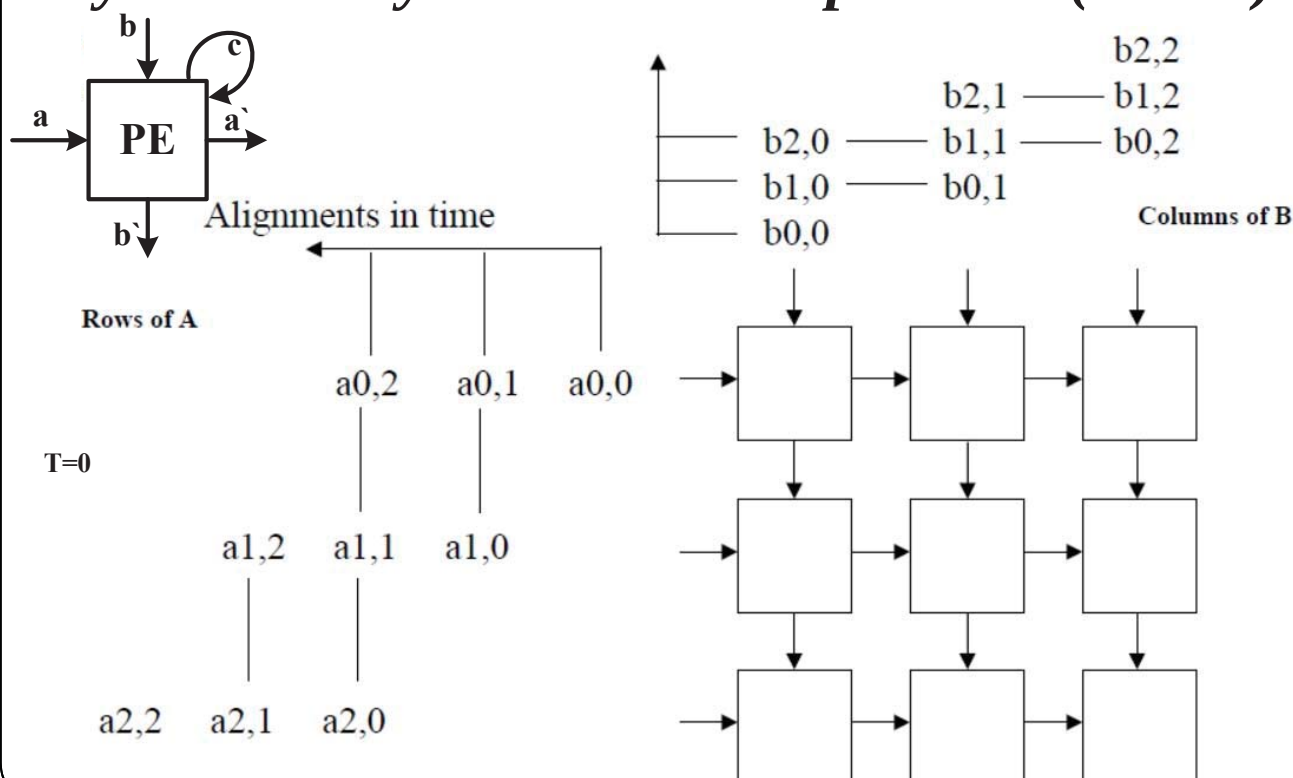
- i : the row index
- j : the column index

■ Signal flow

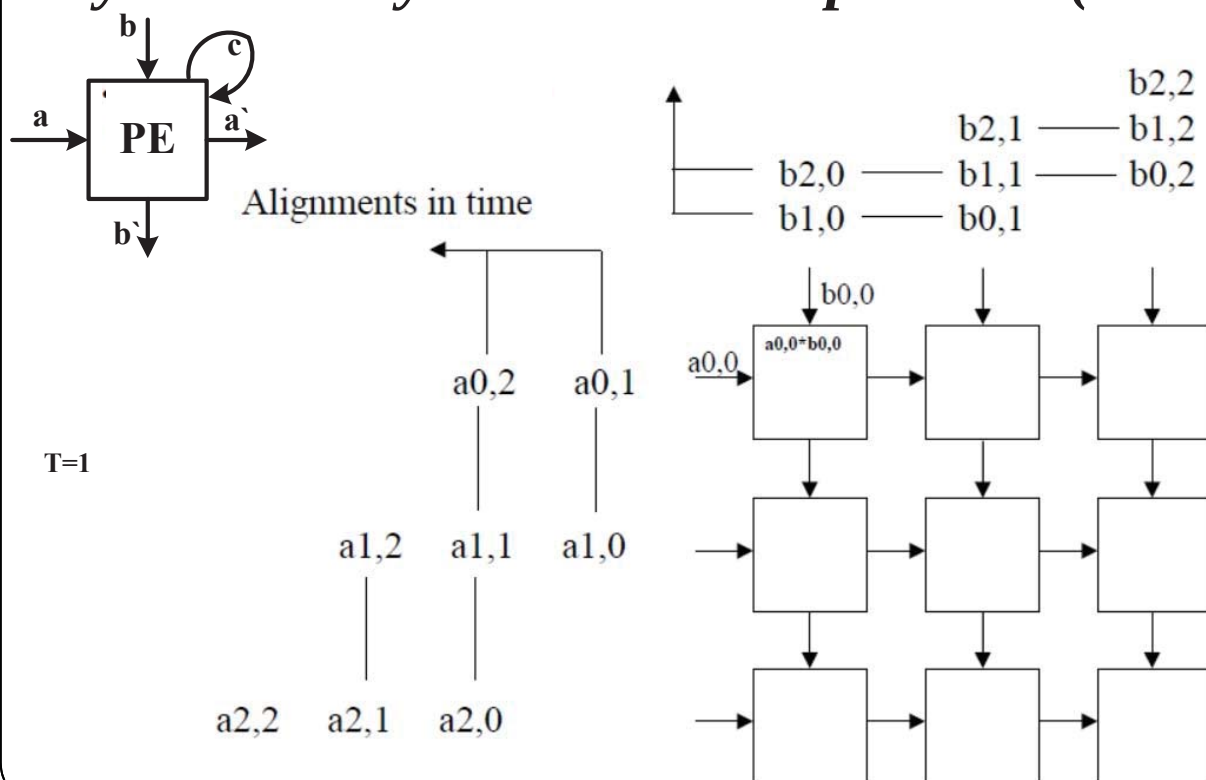
- $a'_{ij} = a_{ij}$
- $b'_{ji} = b_{ji}$
- $c = c + a_{ij} \cdot b_{ji}$



Systolic Array – Matrix Multiplication (cont'd)



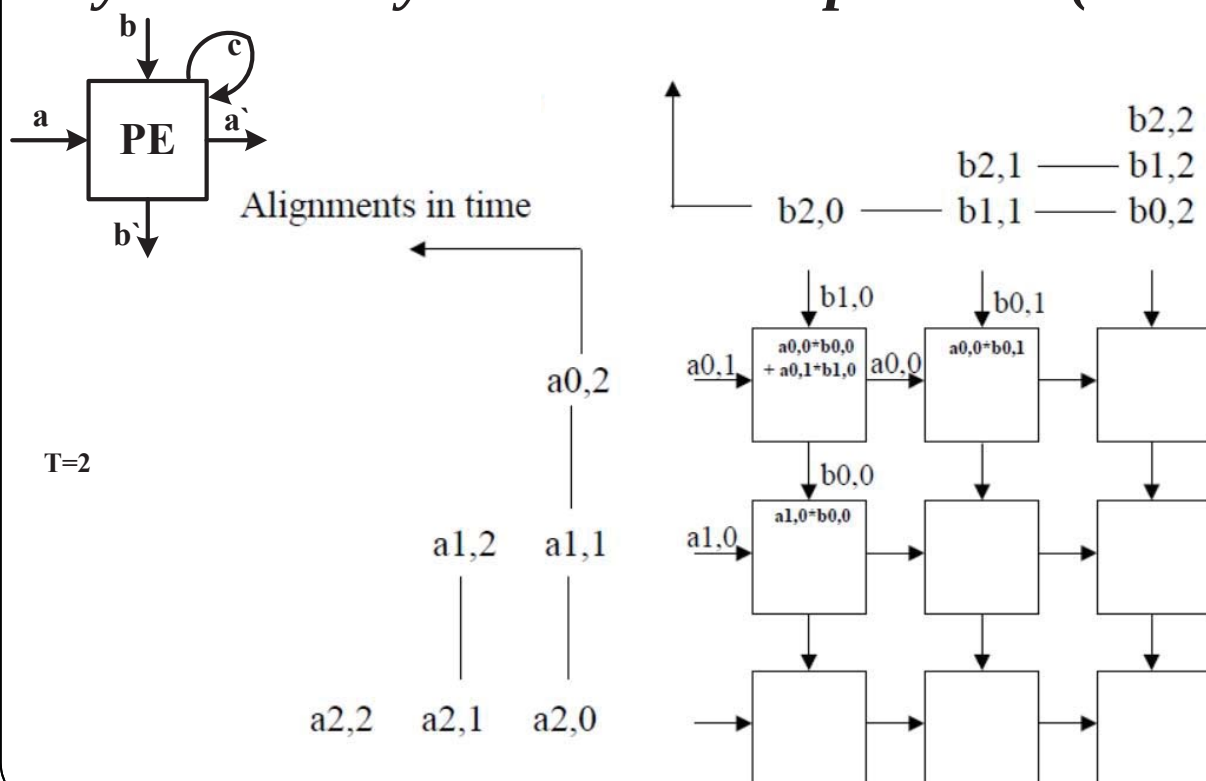
Systolic Array – Matrix Multiplication (cont'd)



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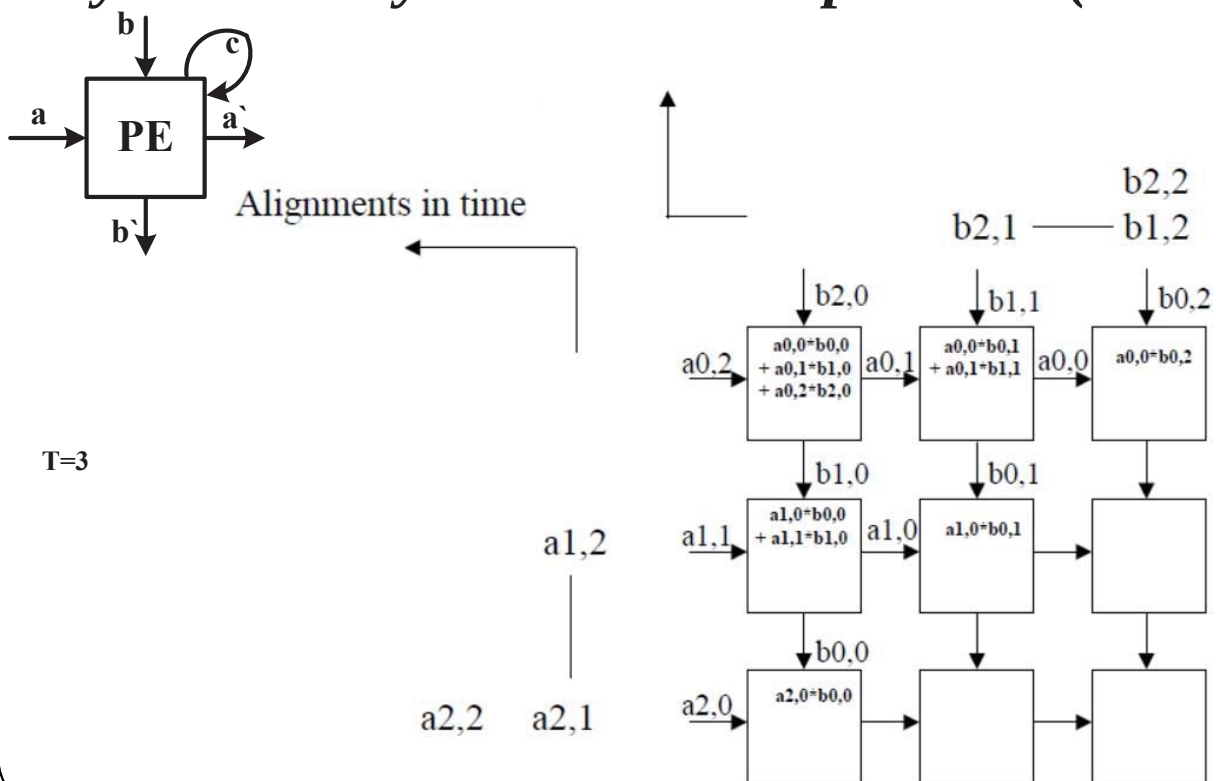
Systolic Array – Matrix Multiplication (cont'd)



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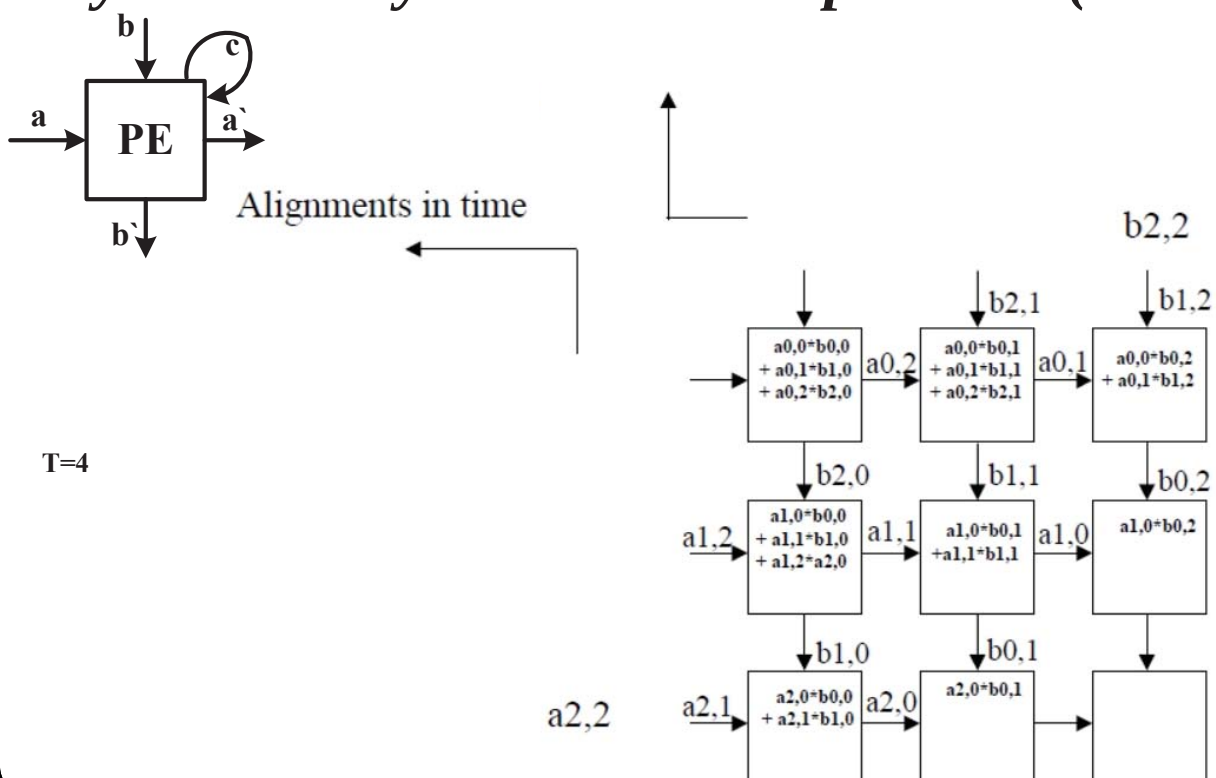
Systolic Array – Matrix Multiplication (cont'd)



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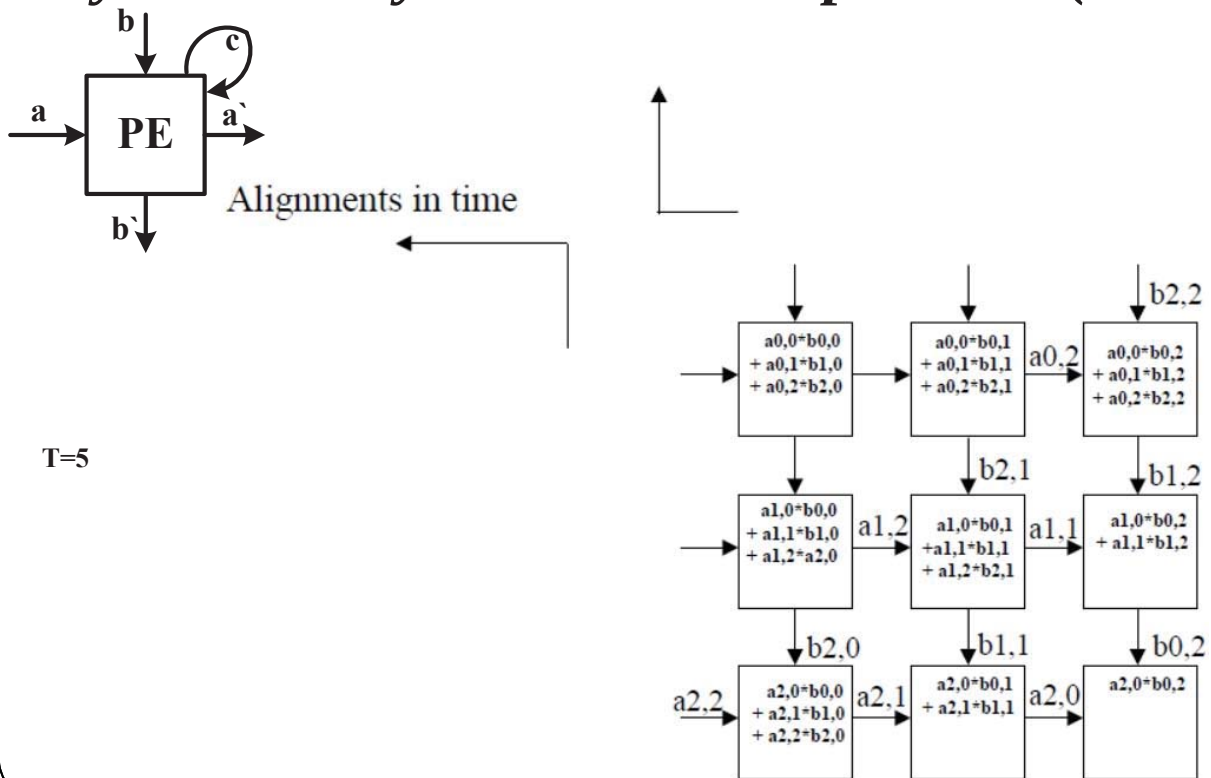
Systolic Array – Matrix Multiplication (cont'd)



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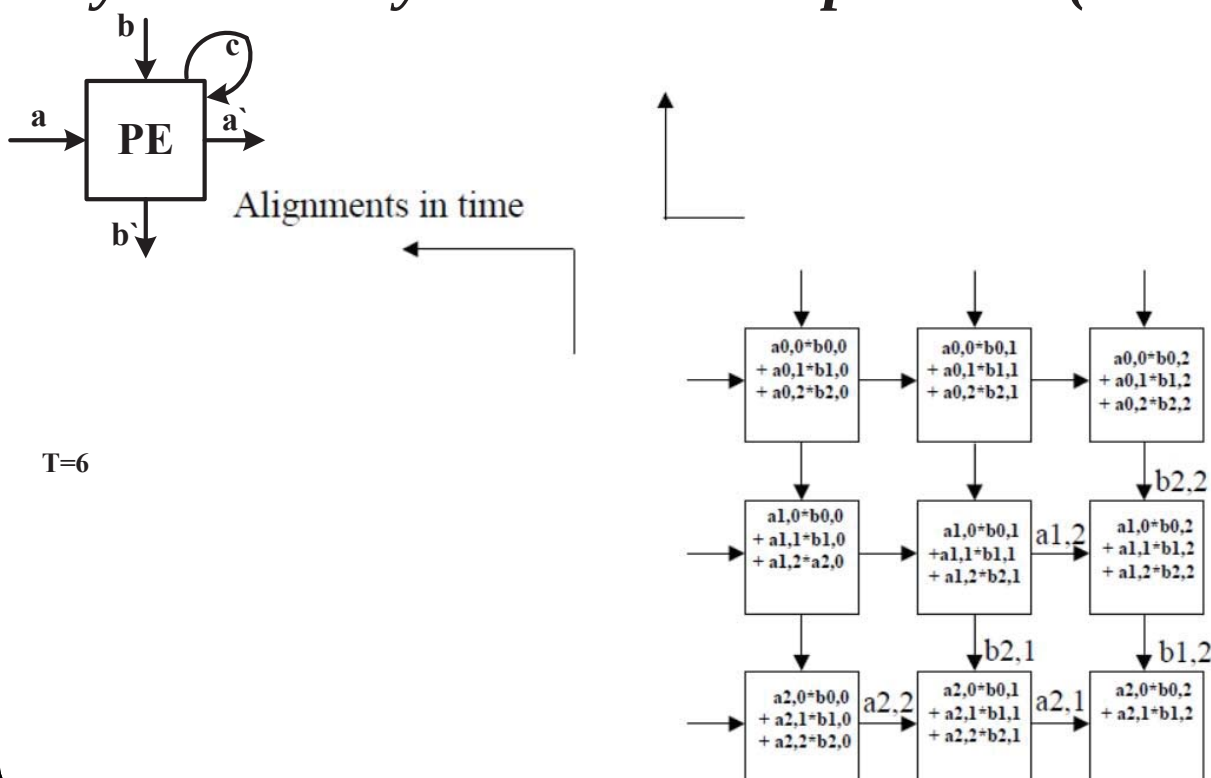
Systolic Array – Matrix Multiplication (cont'd)



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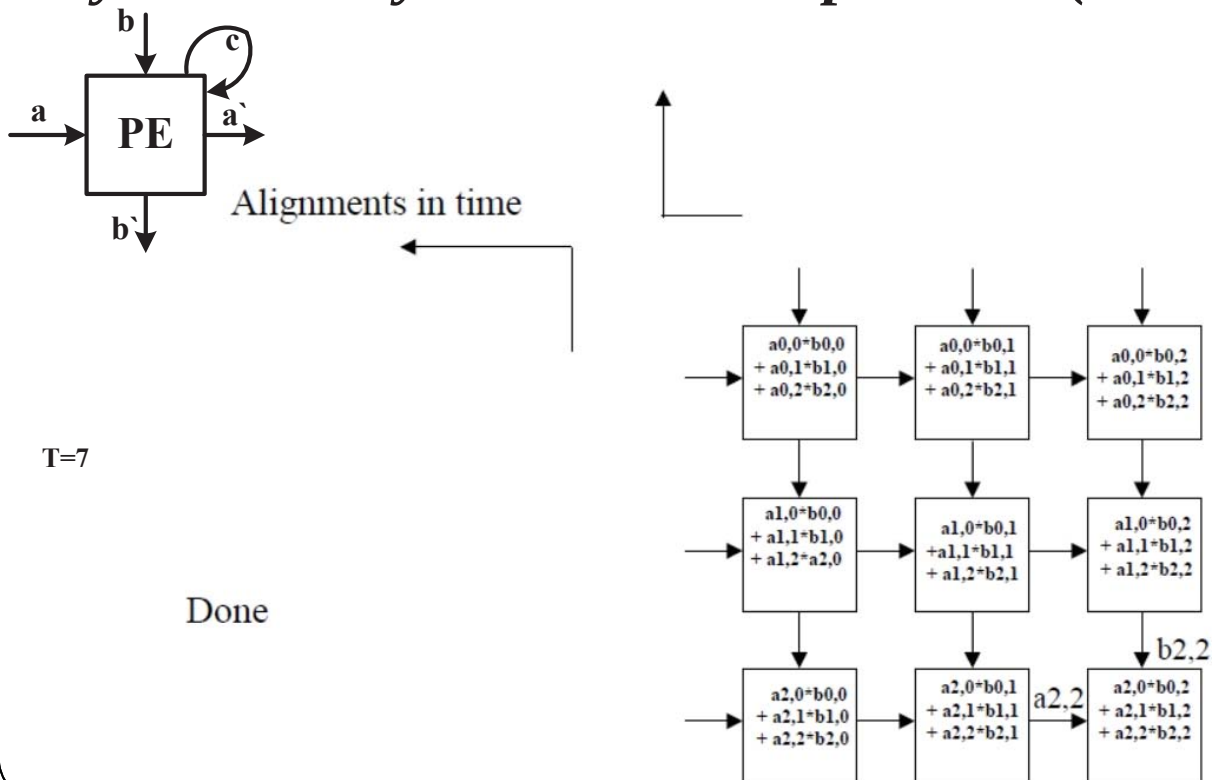
Systolic Array – Matrix Multiplication (cont'd)



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Systolic Array – Matrix Multiplication (cont'd)



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Systolic Convolution Array

□ Type-1: Convolution array

$$\blacksquare y_i = w_1 x_i + w_2 x_{i+1} + \dots$$

■ Processing element (PE)

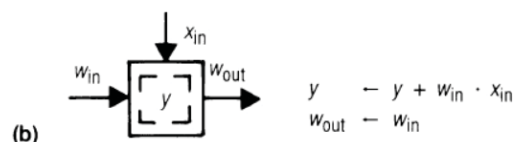
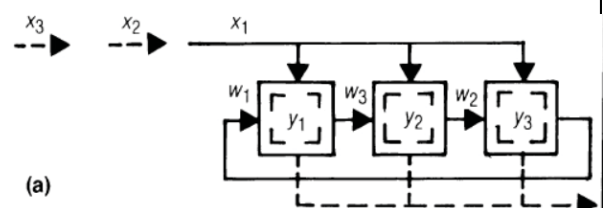
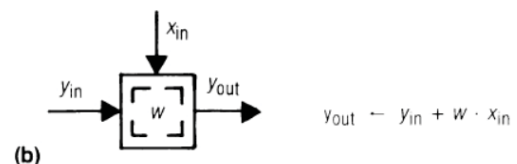
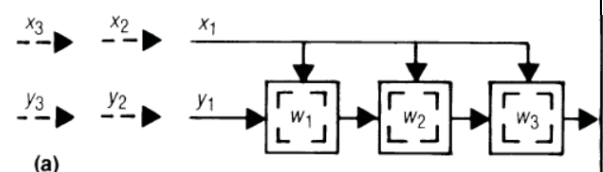
$$y_{out} = y_{in} + w_1 x_i$$

□ Type-2: Convolution array

■ Processing element (PE)

$$y_{out} = y_{in} + w_{in} x_i$$

$$w_{out} = w_{in}$$



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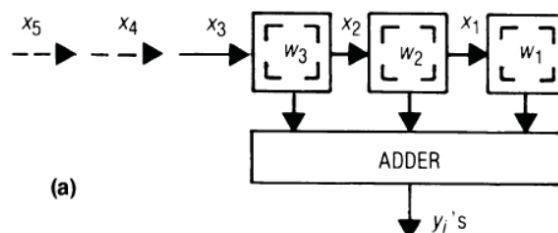
Systolic Convolution Array (cont'd)

□ Type-3: Convolution array

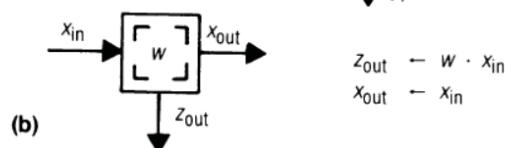
■ PE

$$Z_{out} = W \cdot x_i$$

$$x_{out} = x_{in}$$



(a)



(b)

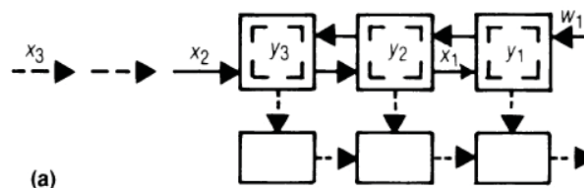
□ Type-4: Convolution array

■ PE

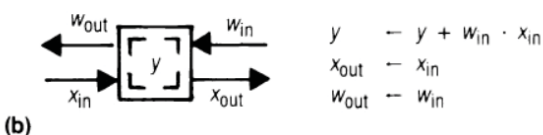
$$y = y + w_{in}x_i$$

$$x_{out} = x_{in}$$

$$w_{out} = w_{in}$$



(a)



(b)

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Systolic Convolution Array (cont'd)

□ Type-5: Convolution array

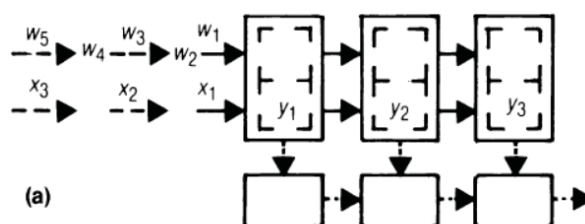
■ PE

$$y = y + w_{in}x_i$$

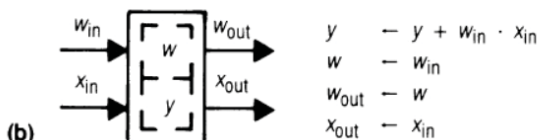
$$w = w_{in}$$

$$w_{out} = w$$

$$x_{out} = x_{in}$$



(a)



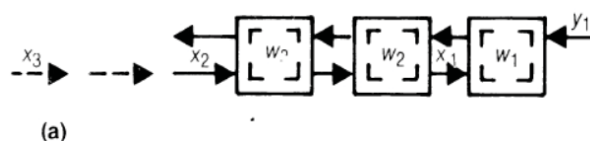
(b)

□ Type-6: Convolution array

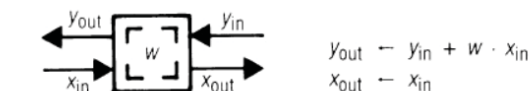
■ PE

$$y_{out} = y_{in} + w \cdot x_{in}$$

$$x_{out} = x_{in}$$



(a)



(b)

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Systolic Convolution Array (cont'd)

□ Type-6: Convolution array (cont'd)

■ PE

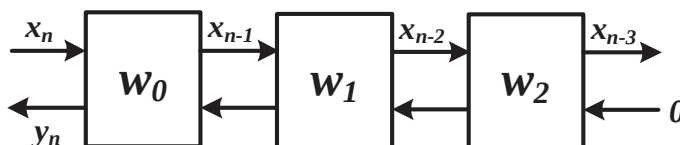
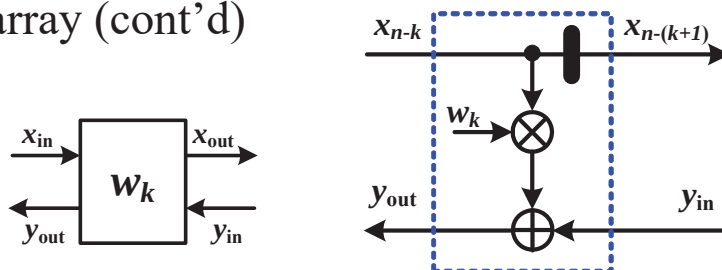
$$y_{out} = y_{in} + w_k x_{in}$$

$$x_{out} = x_{in}$$

$$y_n = \sum_{k=0}^{N-1} w_k w_{n-k}$$

■ $N=3$

$$y_n = \sum_{k=0}^2 w_k x_{n-k}$$



Systolic Convolution Array (cont'd)

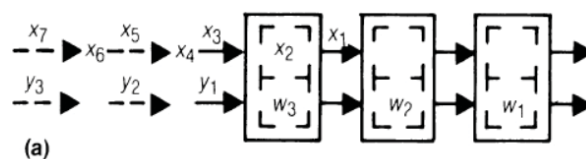
□ Type-7: Convolution array

■ PE

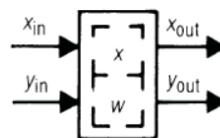
$$y_{out} = y_{in} + w \cdot x_i$$

$$x = x_{in}$$

$$x_{out} = x$$



(a)



(b)

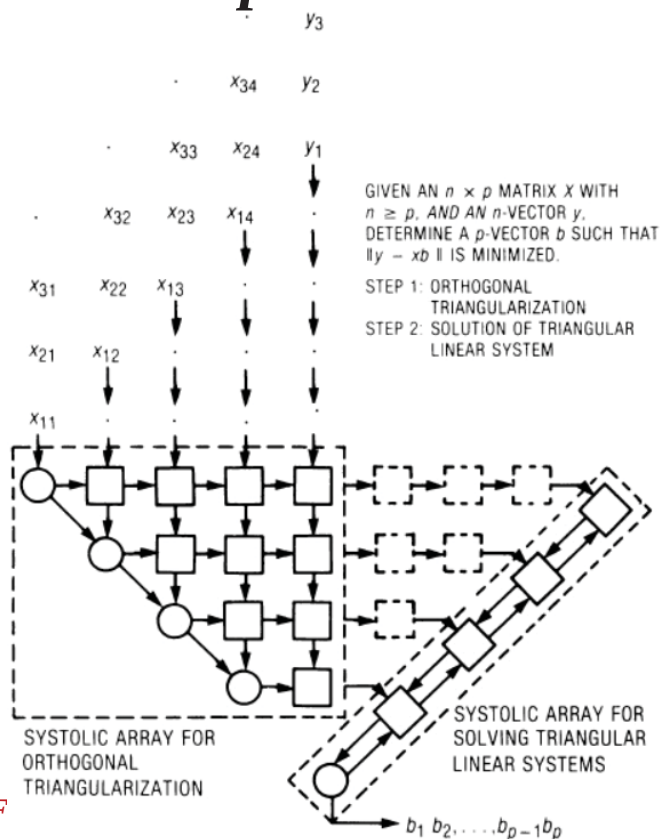
$$y_{out} = y_{in} + w \cdot x_{in}$$

$$x = x_{in}$$

$$x_{out} = x$$

Systolic Array – Matrix Operation

□ Triangular linear systems



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Givens Rotation for QR Decomposition

□ Real Givens rotation

$$\begin{bmatrix} c & s \\ -s & c \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} r \\ 0 \end{bmatrix}$$

$$\theta = \tan^{-1}(b/a) \text{ and } r = \sqrt{a^2 + b^2}$$

□ Complex Givens rotation (two angles rotation)

$$\begin{bmatrix} c_1 & s_1 e^{j\theta_2} \\ -s_1 e^{-j\theta_2} & c_1 \end{bmatrix} \begin{bmatrix} a_1 + ja_2 \\ b_1 + jb_2 \end{bmatrix} = \begin{bmatrix} r_1 + jr_2 \\ 0 \end{bmatrix}$$

$$\theta_a = \tan^{-1}(a_2/a_1) \text{ and } R_a = \sqrt{a_1^2 + a_2^2}$$

$$\theta_b = \tan^{-1}(b_2/b_1) \text{ and } R_b = \sqrt{b_1^2 + b_2^2}$$

$$\theta_1 = \tan^{-1}(R_b/R_a) \text{ and } \theta_2 = \theta_a - \theta_b$$

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Givens Rotation for QR Decomposition (cont'd)

□ Three angles complex rotation (TACR)

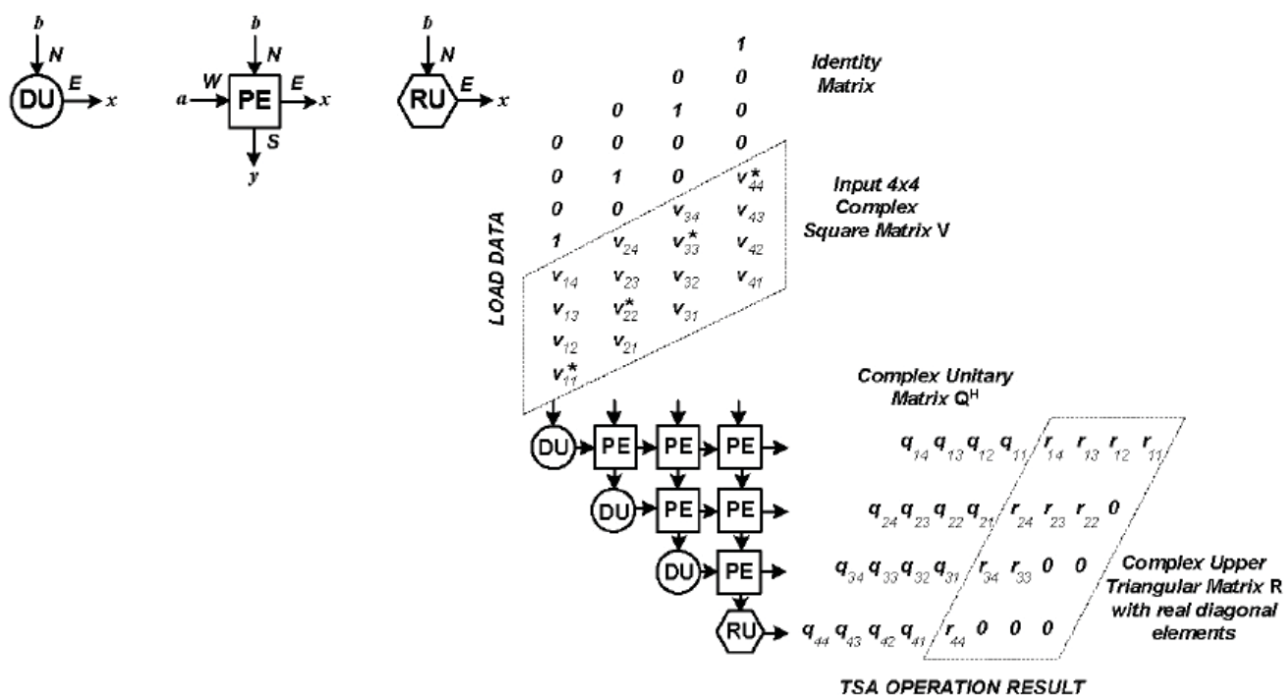
$$\begin{bmatrix} c_1 e^{j\theta_2} & s_1 e^{j\theta_3} \\ -s_1 e^{j\theta_2} & c_1 e^{j\theta_3} \end{bmatrix} \begin{bmatrix} a_1 + ja_2 \\ b_1 + jb_2 \end{bmatrix} = \begin{bmatrix} R \\ 0 \end{bmatrix}$$

$$\blacksquare \theta_1 = \tan^{-1}(R_b/R_a)$$

$$\blacksquare \theta_2 = -\theta_a$$

$$\blacksquare \theta_3 = -\theta_b$$

QR Decomposition



QR Decomposition (cont'd)

□ Real-value QRD: $H = QR$

$$\blacksquare H = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{31} & a_{33} \end{bmatrix} \rightarrow R_{23}R_{13}R_{12}H = R$$

$$\blacksquare R_{23}R_{13}R_{12}H = Q^H H = R \rightarrow H = QR$$

$$\blacksquare \text{Step-1: } \begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} c_{12}a_{11} + s_{12}a_{12} & c_{12}a_{12} + s_{12}a_{22} & c_{12}a_{13} + s_{12}a_{23} \\ -s_{12}a_{11} + c_{12}a_{21} & -s_{12}a_{12} + c_{12}a_{22} & -s_{12}a_{13} + c_{12}a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

QR Decomposition (cont'd)

$$\blacksquare \text{Step-2: } \begin{bmatrix} c_{13} & 0 & s_{13} \\ 0 & 1 & 0 \\ -s_{13} & 0 & c_{13} \end{bmatrix} \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix}$$

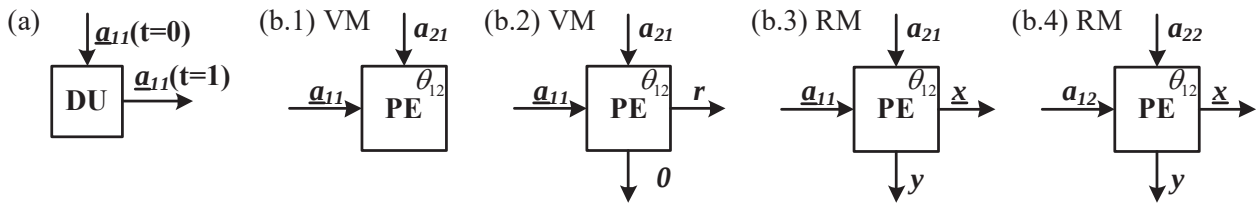
$$\rightarrow \begin{bmatrix} c_{13}a'_{11} + s_{13}a'_{13} & c_{13}a'_{12} + s_{13}a'_{23} & c_{13}a'_{13} + s_{13}a_{33} \\ 0 & a'_{22} & a'_{23} \\ -s_{13}a'_{11} + c_{13}a_{31} & -s_{13}a'_{12} + c_{13}a_{32} & -s_{13}a'_{13} + c_{13}a_{33} \end{bmatrix}$$

$$\blacksquare \text{Step-3: } \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{bmatrix} \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix} = \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a''_{22} & a''_{23} \\ 0 & 0 & a''_{33} \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & c_{23}a'_{22} + s_{23}a'_{32} & c_{23}a'_{23} + s_{23}a'_{33} \\ 0 & -s_{23}a'_{22} + c_{23}a'_{32} & -s_{23}a'_{23} + c_{23}a'_{33} \end{bmatrix}$$

Triangular Systolic Array

□ DU and PE



■ (a) Delay unit

■ (b) CORDIC in PE

- (b.1) PE in VM to find $\theta_{12} = \tan^{-1}(a_{21}/a_{11})$ and $r = \sqrt{a_{11}^2 + a_{21}^2}$
- (b.2) PE in VM to make a_{21} zero and $a'_{11} = r$ (Using VM instead of RM)
- (b.3) PE in RM to realize rotation and bypass the control signal from a_{11} to x
- (b.4) PE in RM to realize rotation and bypass the control signal from a_{22} to y

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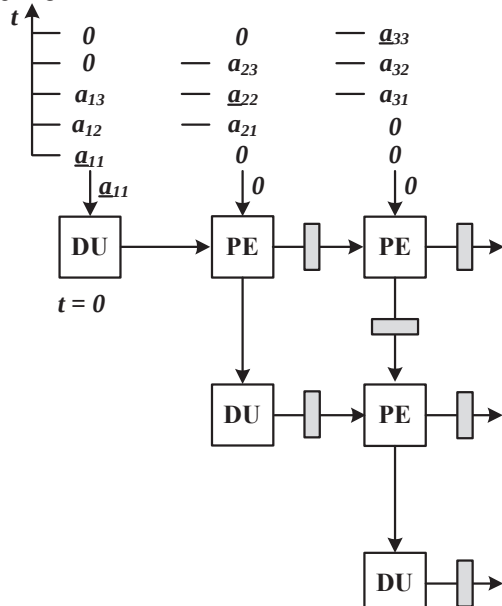
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Triangular Systolic Array (cont'd)

□ Real-value QRD: $H = QR$

$$\begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

□ $t=0$



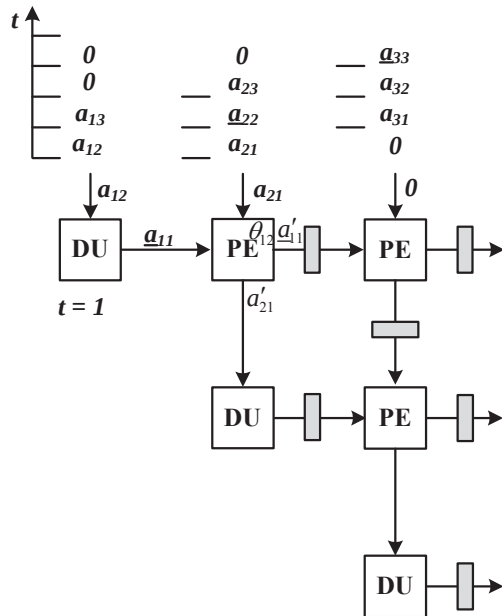
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Triangular Systolic Array (cont'd)

□ $t=1$

$$\begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$



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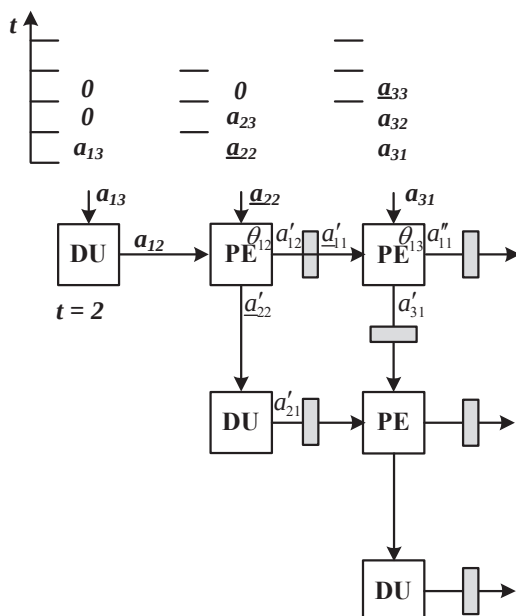
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Triangular Systolic Array (cont'd)

□ $t=2$

$$\begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{bmatrix} c_{13} & 0 & s_{13} \\ 0 & 1 & 0 \\ -s_{13} & 0 & c_{13} \end{bmatrix} \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix}$$

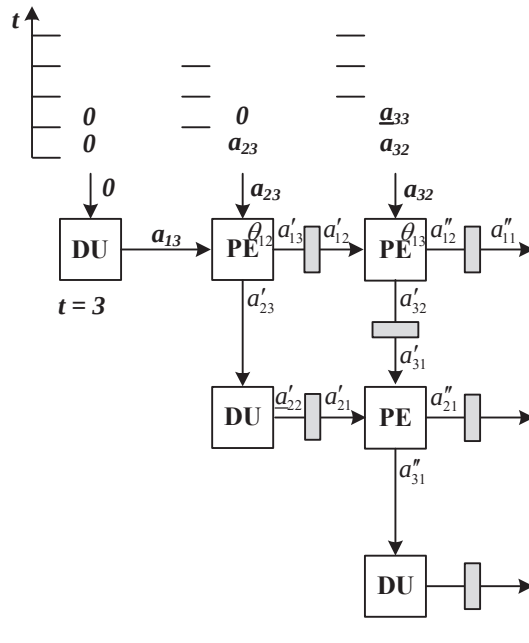


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Triangular Systolic Array (cont'd)

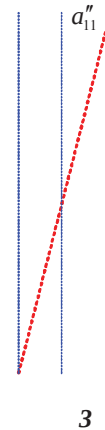
□ $t=3$



$$\begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{bmatrix} c_{13} & 0 & s_{13} \\ 0 & 1 & 0 \\ -s_{13} & 0 & c_{13} \end{bmatrix} \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{bmatrix} \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix} = \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a'_{33} \end{bmatrix}$$

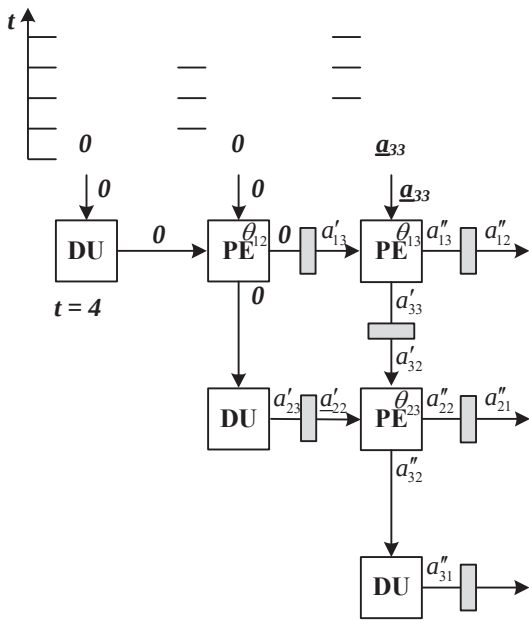


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Triangular Systolic Array (cont'd)

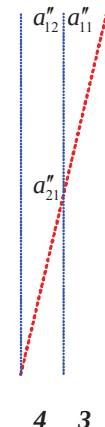
□ $t=4$



$$\begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{bmatrix} c_{13} & 0 & s_{13} \\ 0 & 1 & 0 \\ -s_{13} & 0 & c_{13} \end{bmatrix} \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{bmatrix} \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix} = \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a'_{33} \end{bmatrix}$$

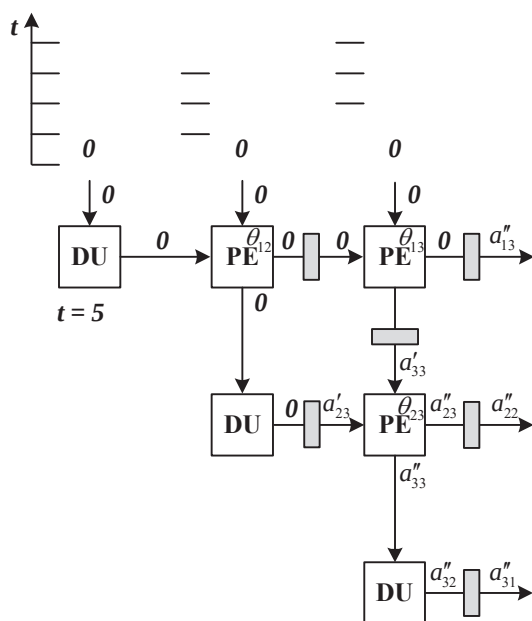


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Triangular Systolic Array (cont'd)

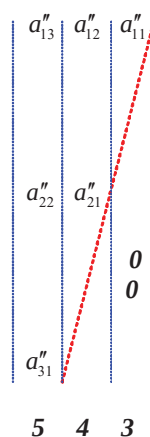
□ $t=5$



$$\begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{bmatrix} c_{13} & 0 & s_{13} \\ 0 & 1 & 0 \\ -s_{13} & 0 & c_{13} \end{bmatrix} \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{bmatrix} \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix} = \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a''_{22} & a''_{23} \\ 0 & 0 & a''_{33} \end{bmatrix}$$

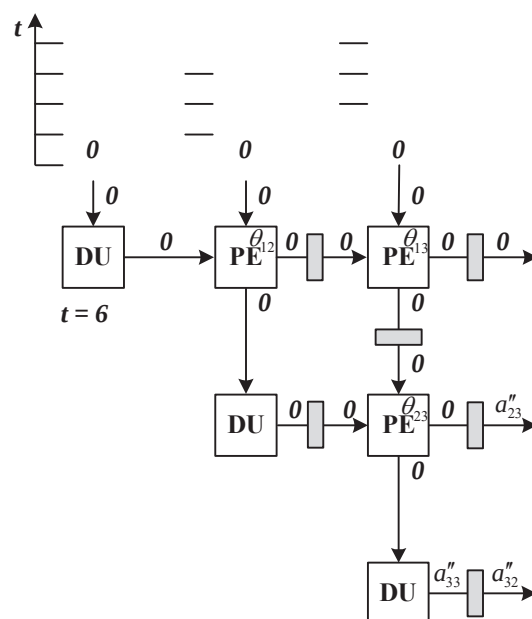


$t =$
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Triangular Systolic Array (cont'd)

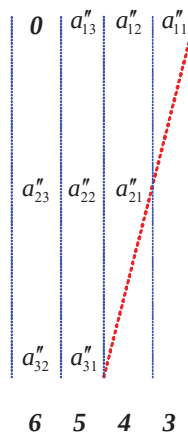
□ $t=6$



$$\begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{bmatrix} c_{13} & 0 & s_{13} \\ 0 & 1 & 0 \\ -s_{13} & 0 & c_{13} \end{bmatrix} \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{bmatrix} \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix} = \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a''_{22} & a''_{23} \\ 0 & 0 & a''_{33} \end{bmatrix}$$

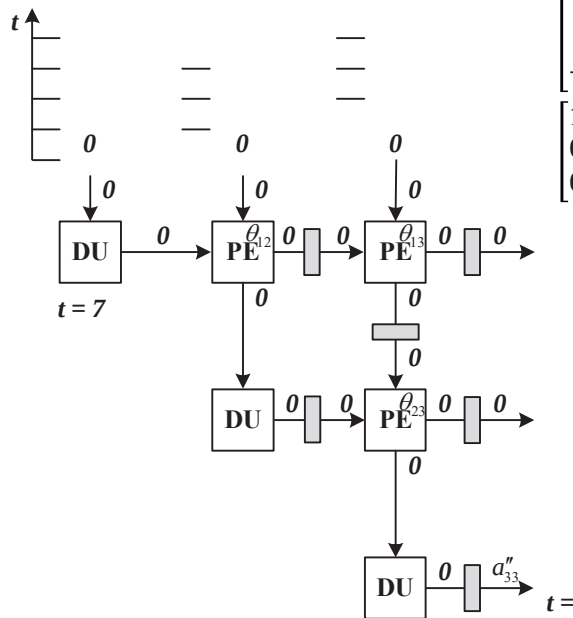


$t =$
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Triangular Systolic Array (cont'd)

□ $t=7$



$$\begin{bmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{bmatrix} c_{13} & 0 & s_{13} \\ 0 & 1 & 0 \\ -s_{13} & 0 & c_{13} \end{bmatrix} \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{bmatrix} \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix} = \begin{bmatrix} a''_{11} & a''_{12} & a''_{13} \\ 0 & a''_{22} & a''_{23} \\ 0 & 0 & a''_{33} \end{bmatrix}$$

